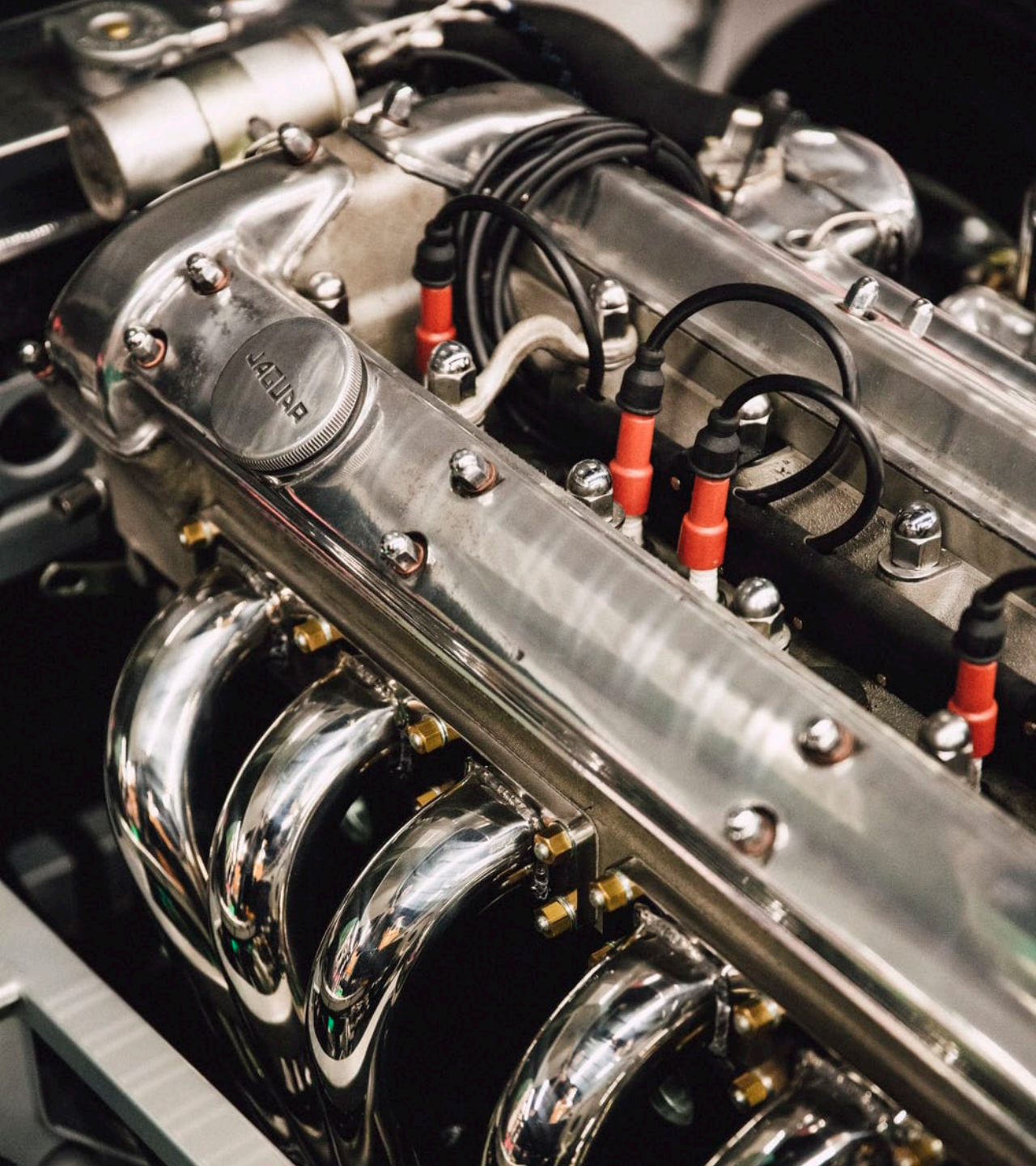




ME 300:
Engineering Thermodynamics

Processes and Cycles

Thermodynamic process

Process : state 1 $\rightarrow$ state 2

Types of thermodynamic processes

Isothermal $\rightarrow$ const. T

Isobaric $\rightarrow$ const. P

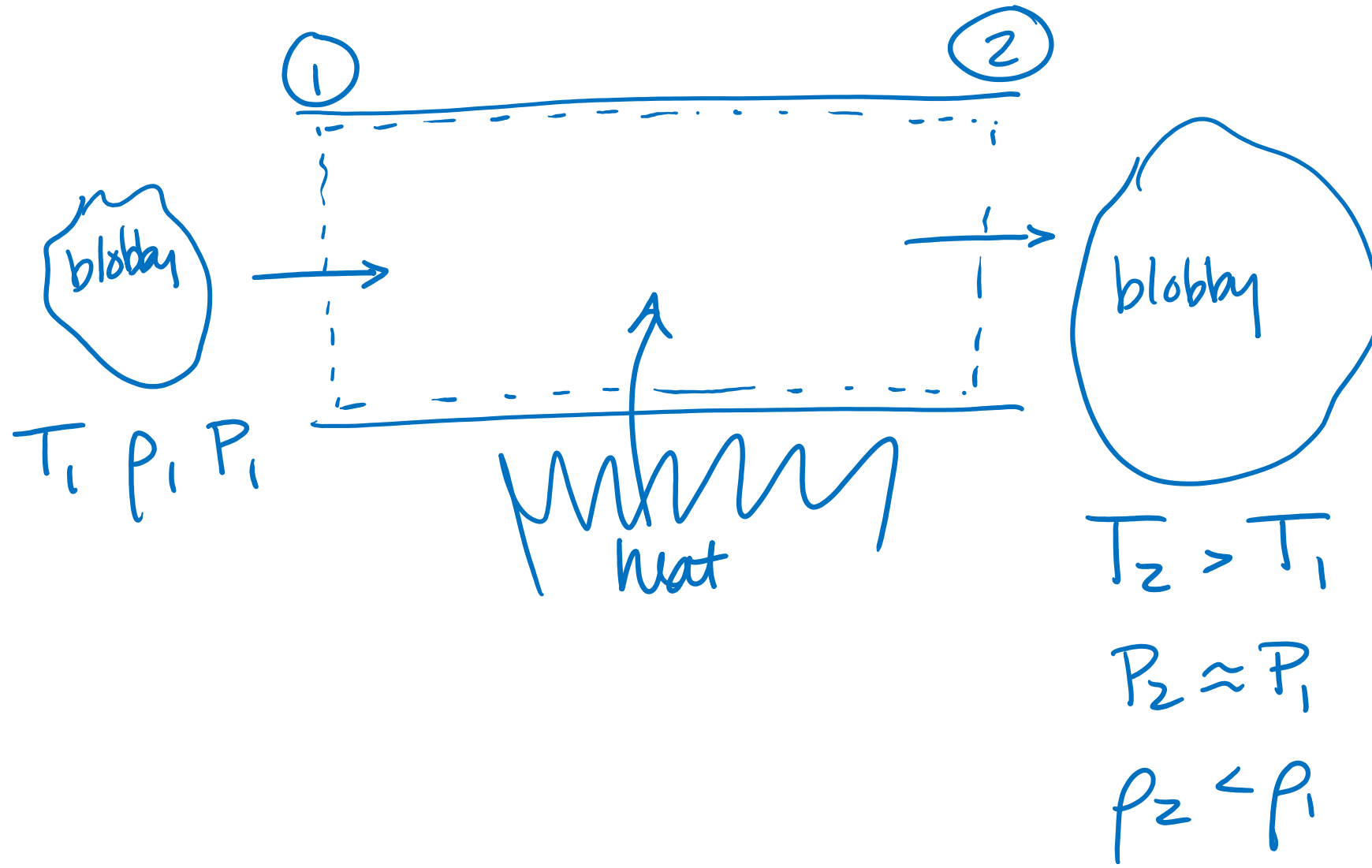
Isochoric $\rightarrow$ const. V

Isentropic $\rightarrow$ const. entropy (S)

adiabatic $\rightarrow$ no heat transfer

passive $\rightarrow$ no work

Flow processes



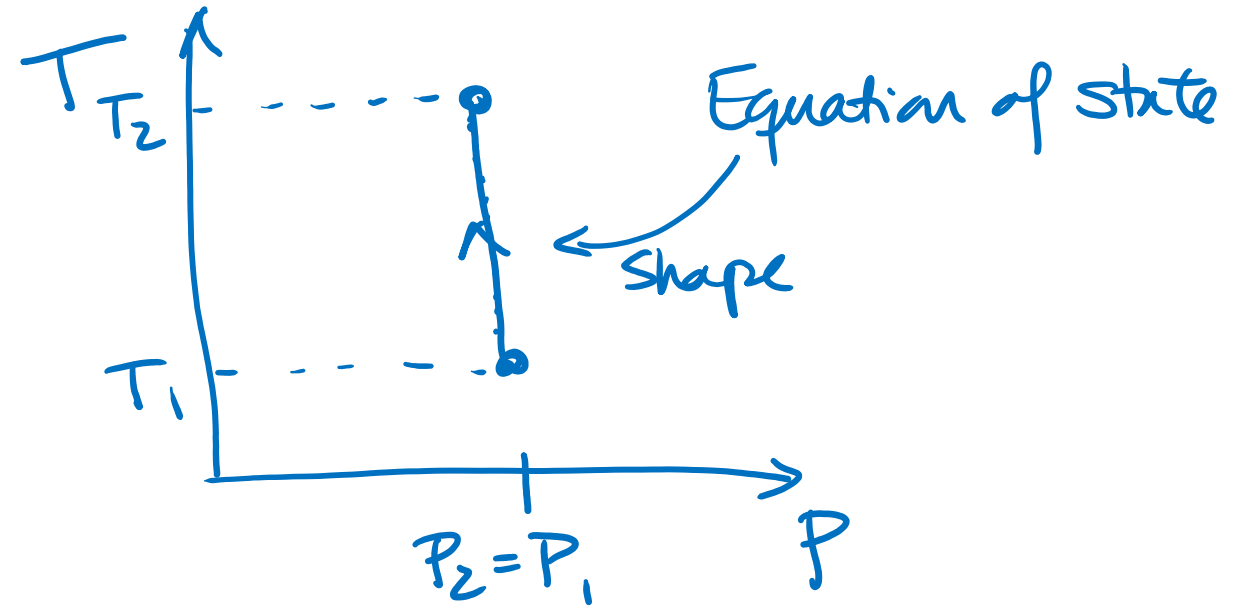
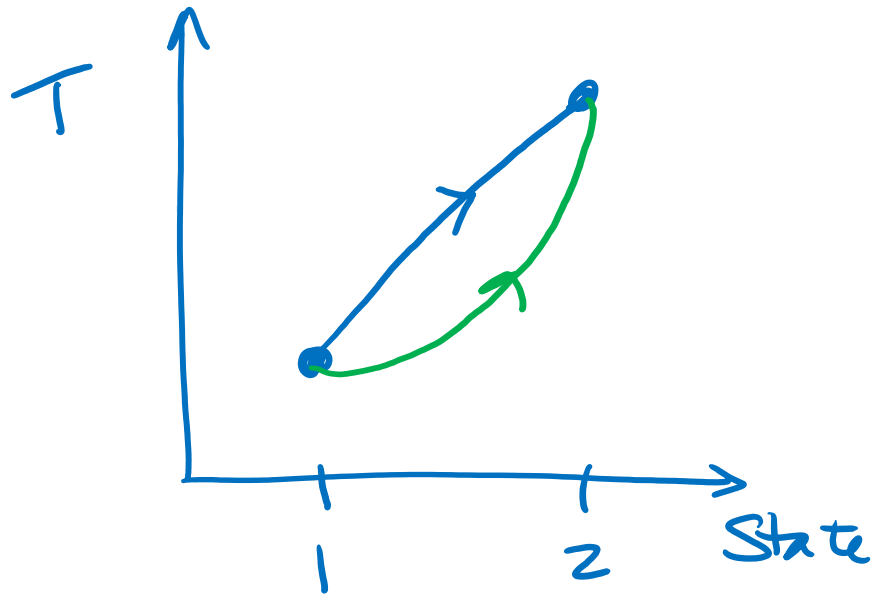


What happens to
blobby as it goes
through the
furnace?

Flow processes

Thermodynamic path

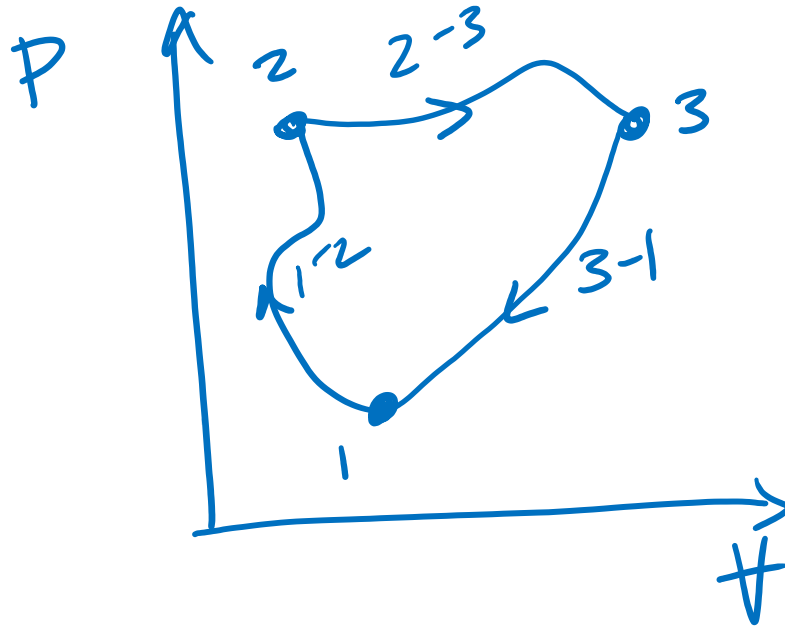
Path: succession of thermodynamic states a system passes through during a process



Thermodynamic cycle

Cycle: sequence of processes by which working fluid returns to its original state

↑
inside system
G+



Foreshadowing: Thermodynamic cycles

Cycles $\rightarrow$ system returns to its original state



surrounding X